

MENTIX-Hub

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Project Overview

Introduction

MENTIX-Hub is a platform for project sharing, mentorship, and collaboration, helping students showcase projects, receive guidance, and connect with others. The name MENTIX comes from Mentor + Interaction + Exchange.

Website Overview

We separate Ui into 2 sides is admin and client side:

- **Client side:** 2 roles (Student, Mentor)
- **Admin side:** 3 roles (Super admin, Developer admin, moderator admin.)

RM - Clinent

User role

1. Mentor
2. Studdent

RM - Admin

User Privilege & Access control

Database User:

DB User	Privileges	Why
mentix_user	SELECT, INSERT, UPDATE, DELETE on mentix_hub.*	Normal app CRUD only
mentix_dev	Above + LOCK TABLES, SHOW VIEW, TRIGGER	Admin & backup operations
mentix_root	ALL PRIVILEGES + WITH GRANT OPTION on mentix_hub.*	Schema changes, restore

Finger 1. create connectio pool for mentix user

```
const userDB = mysql.createPool({
  ...baseConfig,
  user: config.db.users.user.user,
  password: config.db.users.user.password,
  connectionLimit: 15
});
```

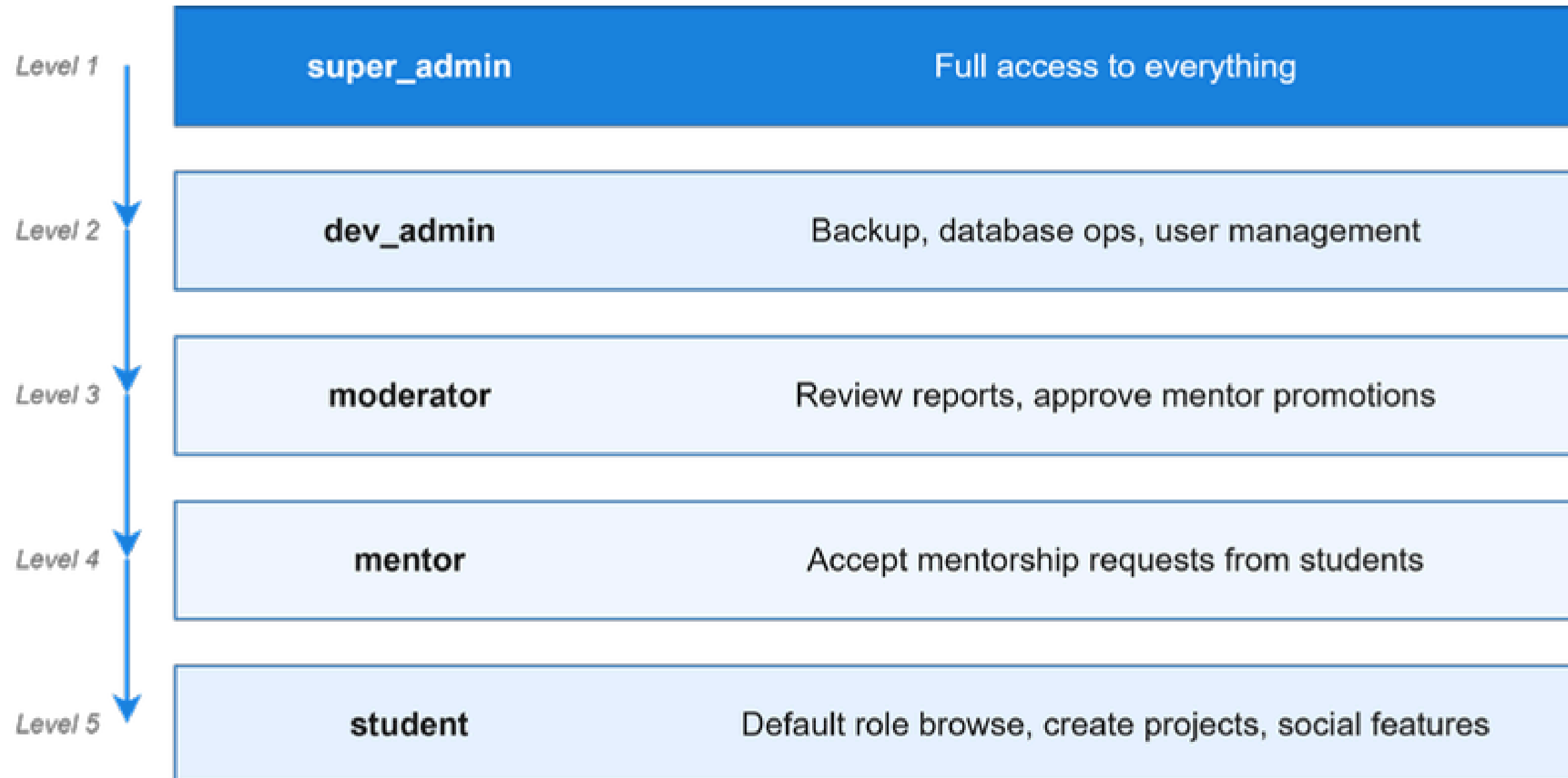
Finger 2. create connectio pool for mentix dev

```
const devDB = mysql.createPool({
  ...baseConfig,
  user: config.db.users.dev.user,
  password: config.db.users.dev.password,
  connectionLimit: 5,
  multipleStatements: true
});
```

Finger 3. create connectio pool for mentix root

```
const rootDB = mysql.createPool({
  ...baseConfig,
  user: config.db.users.root.user,
  password: config.db.users.root.password,
  connectionLimit: 3,
  multipleStatements: true
});
```

Application Users:



Query Optimization

Query optimization is the process of improving SQL queries to make them execute faster and use fewer database resources, resulting in better database performance.

Query Optimization Use in MENTIX_Hub system:

- 1. Connnection Pool**
- 2. Indexing Strategy**
- 3. Parallel Execution & Pagination**

Connection Pool

Instead of creating a new database connection for every request, the system reuses existing connections from a pool, making it faster and more efficient.

Benefits:

- Faster response by reusing connections.
- Prevents too many database connections.
- Improves security with separate access levels.
- Keeps connections active for stable cloud database access.

The connection pool example is on finger 1. , finger 2., finger 3.

Indexing Strategy

Index Summary

Index Type	Count	Code Example	Purpose
Cluster	21	id INT PRIMARY KEY AUTO_INCREMENT on all tables; PRIMARY KEY (user_id, project_id) on hearts, favorites	Row data stored sorted by PK. Fastest row access
B-Tree Non-Cluster	42	INDEX idx_author_id (author_id) on projects; INDEX idx_status (status) on reports; INDEX idx_user_type (user_id, type) on email_verifications; INDEX idx_expires_at (expires_at) on refresh_tokens	Speeds up JOINS, WHERE filters, ORDER BY, and uniqueness checks
FULLTEXT	1	FULLTEXT INDEX ft_title_description (title, description) on projects	Keyword search on text columns

Indexes reduce the search time from approximately $O(n)$ to $O(\log n)$ by using a B-Tree structure instead of scanning every row.

Parallel Execution & Pagination

Parallel Execution: Executes multiple independent queries simultaneously. We implement in Node.js using Promise.all().

Specific selection: Retrieves only the required columns instead of using SELECT *.

Pagination: Retrieves data in smaller batches using LIMIT and OFFSET. it reduces query execution time and memory usage.

Example:

```
SELECT h.*, p.title, p.description, p.thumbnail, p.view_count, p.created_at,  
       u.username, u.full_name, u.avatar_url  
FROM hearts h  
JOIN projects p ON h.project_id = p.id  
JOIN users u ON p.author_id = u.id  
WHERE h.user_id = ?  
ORDER BY h.created_at DESC  
LIMIT ? OFFSET ?  
;
```

Backup & Recovery

Backup

- Type of backup :
Manual
Scheduled
- Format of file backup:
sql script
CSV

Storage file Backup

Source file backup store in the Cloudinary

Recovery

- Only Super Admin can restore backups.
- Display only successful backups.
- Download the selected backup from Cloudinary.
- Restore the database from the SQL file.
- Clean up temporary files after completion.

Conclusion

For MENTIX-Hub, we set up user privileges to keep data secure, used connection pooling and indexing to speed up queries, and built a backup and recovery system to protect our data. These steps make the platform secure, fast, and reliable.





THANK
YOU!